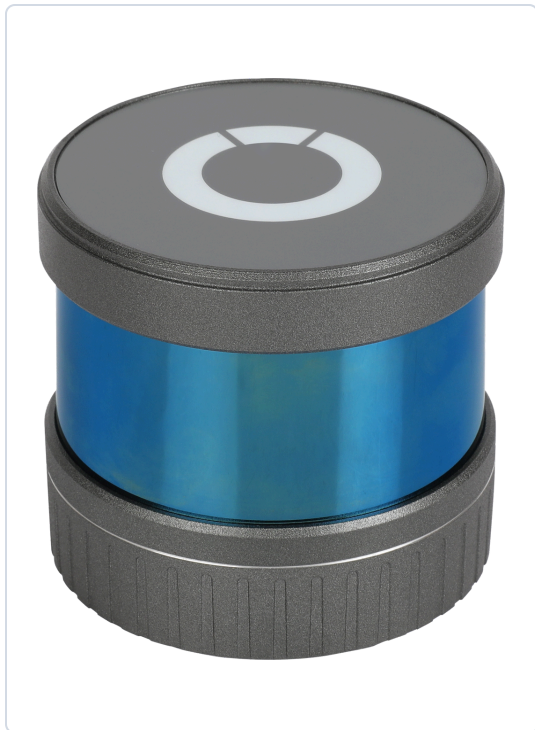


3D 360° MULTI-LINE LIDAR · LR-16F

SIXTEEN LINES OLEI LR-16F AT 100 METRES

A 16-line 3D scanner that sees 100 metres across a full circle in an 800-gram, 97 mm can. 310 000 points a second with timestamps and calibrated reflectivity, and GPS on board for mapping work.



■ WHY THIS SENSOR

- **100 m at 80% reflectivity, 45 m at 10%** — long enough for outdoor mapping and highway-speed perception
- **16 lines across ±15° vertical, 360° horizontal** — a 3D volume from one rotating sensor, not a single plane
- **310 000 points per second** — 0.18° horizontal at 10 Hz, 2° vertical spacing
- **≤ ±5 cm absolute, ≤ ±3 cm repeat accuracy** — tight enough to localise on, over the full 100 m
- **UDP packets carry distance, reflectivity, angle and timestamp** — synchronised time stamps mean the cloud fuses cleanly with IMU and odometry
- **GPS on board** — for georeferenced capture without an external receiver
- **IP66, -10 to +50 °C, ≤ 10 W at 12–30 V DC** — specified for stability and accuracy at temperature extremes
- **97 mm diameter × 85.2 mm, ~800 g** — compact enough to sit on a service robot, not just a survey vehicle

100 m

AT 80% REFLECTIVITY

16

LASER CHANNELS

310K

POINTS / SECOND

360° × 30°

FIELD OF VIEW

■ APPLICATION SCENARIOS



AMR & AGV

Full-circle obstacle avoidance from one chassis mount



SURVEY & MAPPING

360° natural-feature and reflector mapping at range



TERRAIN & CORRIDOR

Real-scene 3D of ground, road and rail corridors



AUTONOMOUS VEHICLES

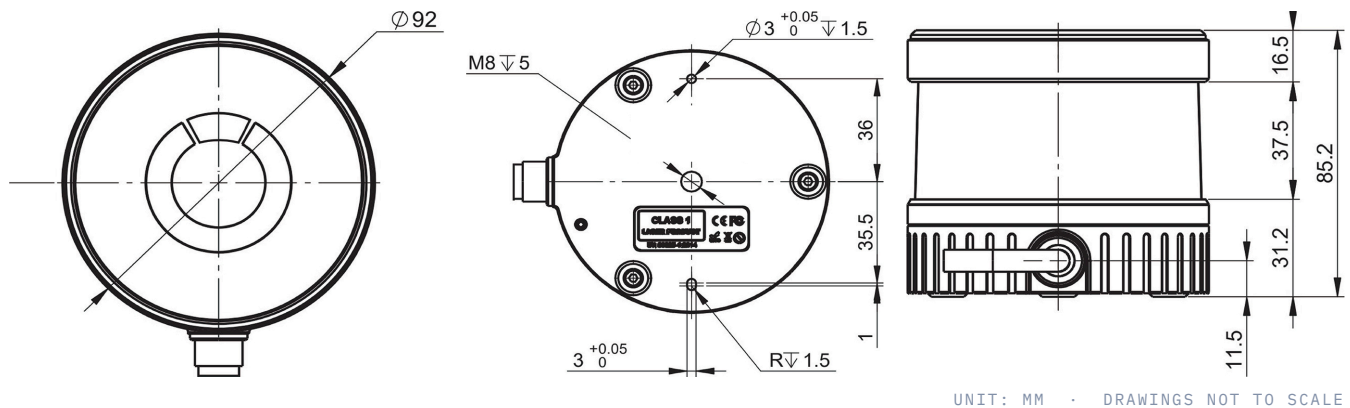
Forward perception for on- and off-road platforms

LR-16F — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Laser wavelength	905 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	16
Returns	Single (strongest, first)
Field of view (vertical)	-15° to +15°
Field of view (horizontal)	360°
Beam divergence	9 mrad longitudinal 3 mrad transversal
Angle resolution (vertical)	2° (no 0° line) ±0.1°
Angle resolution (horizontal)	0.18° ±10% @ 10 Hz
Scanning frequency	5 Hz or 10 Hz
Distance resolution	2 mm
Accuracy	Absolute ≤ ±5 cm Repeat ≤ ±3 cm
Measurement range	0.5–100 m @ 80% reflectivity 0.5–45 m @ 10% · 0.5–100 m @ reflector
Signal strength	0–255
Scanning rate	~310 000 pts/s

PARAMETER	TECHNICAL DATA
Ethernet connection	100 Mbps Ethernet
UDP packets contain	Distances · calibrated reflectivities Rotation angles · synchronised time stamps
GPS	Yes
Connection	12-pin A-coded socket With interface box
Mechanical dimensions	97 × 85.2 mm
Weight	~800 g
Power consumption @ 25 °C	≤ 10 W
Operating voltage	12–30 V DC
Degree of protection	IP66
Operating temperature	-10 to +50 °C
Storage temperature	-20 to +60 °C
Air humidity	95% (non-condensing)
Ambient light limit	< 80 000 lux
Mounting	Bottom
Basic software	OLAMViewer · ROS driver

■ OUTLINE AND MOUNTING



Certification. FCC, EMC, CE, RoHS, REACH. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states no IEC 61496 Type rating and no ISO 13849 performance level. Treat its output as perception and measurement data, not as a rated safety function, and do not use it as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. The 100 m figure is the 80% reflectivity and reflector case; plan on 45 m at 10% reflectivity. Vertical spacing is 2° with no 0° line, so the beam pattern is symmetric about the horizon rather than centred on it — check this against any ground-plane extraction that assumes a centre line. Horizontal resolution is stated at 10 Hz; at 5 Hz it halves.

Installation. Bottom mounting. A 360° sensor needs its full circle clear — mount it above the platform's own structure rather than recessed into it, or the blind sectors will be created by the vehicle instead of the sensor. The 12-pin connection runs through an interface box, so allow space and a cable route for it. OLEI accepts customisation of channel count (4–32), wiring method, explosion-proof construction and protection level — ask MorpheusTEK.

OLEI and the OLEI logo are trademarks of Hangzhou OLE-Systems Co., Ltd. MorpheusTEK is the exclusive North American distributor for OLEI LiDAR. Specifications are subject to change without notice; confirm current values with MorpheusTEK before design-in. Doc. MTK-OLEI-LR16F-2026.09 · Source: OLEI LR-16F 3D LiDAR Sensor datasheet, DOC.# DSEN-16F-202505.